



SHM-X MDU OF TAP CHANGER OPERATING INSTRUCTIONS

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1. Foreword

1.1 Manufacturer :

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1.2.Subject to change without notice

The information contained in this technical file comprise the technical specifications released at the time of printing. Significant modifications will be included in a new edition of the technical file. The document and version numbers for this technical file are shown in the footer

1.3 Safety

Safety instructions are explained in this manual. They must be observed and followed to avoid relevant dangers.

Personal safety equipment must be worn during work to prevent potential risks.

	Protective clothing for protecting personnel from hazards in the work environment.
	Safety shoes for protecting personnel from falling sharp objects and slippery surfaces.
	Goggles for protecting eyes and facial area from impact, dust, liquid splash and high heat hazards.
	Hard hat for protecting head from injury.
	Hearing protection for protecting hearing from damage.
	Protective gloves for protecting hands from mechanical, chemical, and electrical hazards.

Table 1-1 Personal safety equipment list

1.4 Hazard communication system

Warnings in this technical file use the following format:



SIGNAL WORD!

Consequences
Action

The following signal words are used:

Signal words	Hazard level	Consequences of failure to comply
Danger	Immediate threat of danger	Death or serious injury certainly occur Warning
Warning	Probable threat of danger	Death or serious injury apparently occur Warning
Prohibition	Probable dangerous situation	Minor or moderate injury likely occur
Note	Possible dangerous situation	Material damage may occur

Table 1-2 Signal words in safety instructions

Pictograms warn of dangers:

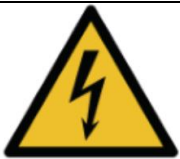
Pictogram	Meaning
	Danger The yellow triangle warning sign means life and health may be at risk if neglecting the instruction. Neglecting this warning can lead to serious or fatal injuries.
	Dangerous electrical voltage
	Fire hazard
	General prohibition sign! The red symbol indicates some actions are prohibited.

Table 1-3 Symbols used in warning notices

1.5 Information system

Information is designed to simplify and improve understanding of particular procedures. In this technical file they are laid out as follows:



NOTE!

The blue note symbol indicates important information that must be noted.

1.6 Safety precautions

Precaution	Action
Electric shock	Always unplug a device before inspection. Stay away from high-voltage equipment and wear protective clothing.
Transformer oil	Collect used transformer oil in dedicated drums. Keep it away from flames or sparks. Do not touch it which may cause damages to health.
Fire	In case of fire, extinguish it with powder, foam or carbon dioxide. Do not use water.
Waste	Wastes are dangerous to the environment or human health. Clean them up in time and use different methods according to the wastes' categories.

Table 1-4 Safety precaution

1.7 Appropriate use

The product and associated equipment and special tools supplied with it comply with the relevant legislation, regulations and standards, particularly health and safety requirements, applicable at the time of delivery.

If used as intended in compliance with the specified requirements and conditions in this technical file as well as the warning notices in this technical file and attached to the product, then the product does not present any hazards for persons, property or the environment. This applies during the entire lifespan, from delivery through installation and operation to disassembly and disposal.

The operational quality assurance system ensures a consistently high quality standard, particularly when it comes to observance of the health and safety requirements.

Use is considered to be appropriate if

- 1) the product is operated according to this technical file, the agreed delivery conditions and technical data, and
- 2) the associated equipment and special tools supplied with it are used solely for the intended purpose and in accordance with the specifications of this technical file.
- 3) the product is used only with the transformer specified in the order.

- 4) the serial number of the manual drive matches that of the SHZV type non-excitation tap switch

1.8 Inappropriate use

Use is considered to be inappropriate if the product is used other than as described in this Chapter 1.7. Shanghai Huaming Power Equipment Co., Ltd does not accept liability for damage from unauthorized or inappropriate changes to the product. Unauthorized changes to the product without consultation with Huaming can lead to personal injury, material damage and operational faults.

1.9 Personnel qualification

The product is designed solely for use in electrical or energy systems and facilities operated by appropriately trained staff. This staff comprises people who are familiar with the installation, assembly, commissioning and operation of such products.

1.10 Operator duty of care

To prevent accidents, faults and damage as well as unacceptable adverse effects on the environment, those responsible for transport, installation, operation, maintenance and disposal of the product or parts of the product must ensure the following:

- All warning and hazard notices are complied with.
- Personnel are instructed regularly in all relevant aspects of operational safety, the Operating Instructions and particularly the safety instructions contained therein.
- Regulations and operating instructions for safe working as well as the relevant instructions for staff procedures in the case of accidents and fires are kept on hand at all times and displayed in the workplace where applicable.
- The product is only used in a sound operational condition and safety equipment in particular is checked regularly for operational reliability.
- Use only replacement parts, lubricants and aids which are authorized by the manufacturer.
- Comply with the specified operating conditions and requirements of the installation location.
- All necessary devices and personal protective equipment for each activity are available.
- The prescribed maintenance intervals and the relevant regulations are complied with.
- Installation, electrical connection and commissioning of the product is only carried out by qualified and trained personnel in accordance with this technical file.
- The operator must ensure appropriate use of the product

2. General

SHM-X operating mechanism (hereinafter referred to as the mechanism) is a new generation of intelligent full-function operating mechanism with independent intellectual property rights of Huaming. It can not only reduce the error rate, but also has strong anti-interference and anti-shock characteristics, and does not have high requirements on the power conditions. At the same time, it has a variety of online monitoring functions to ensure the safety of the switch body and the transformer.



Figure2-1 Overview of SHM-X

3. Main Features of Operating Mechanism

- 1) Asynchronous motors are adapted for all moving parts, and support DC motors are also supported, which can be adapted according to customer requirements. The running accuracy can reach 100% by precise control;
- 2) Replace the traditional signal cable with optical fiber to realize the connection with the remote controller, so can make the installation and usage more accurate and easier;
- 3) The non-contact angle encoder widely used in the automotive industry is adapted, which has strong anti-interference ability and can ensure good performance in various harsh operating environments;
- 4) A variety of online monitoring protections are added in the operating mechanism to ensure the lock and alarm of the tap changer and transformer under abnormal operation conditions, and to display and alarm of the deep oil temperature of the tap changer and transformer;
- 5) It has a locking alarm function for the slow infiltration of the vacuum degree of the tap changer vacuum interrupter and the decrease of the vacuum degree;
- 6) Monitoring and alarm functions for oil-immersed tungsten copper contact tap changers;
- 7) The box cover is made of aluminum alloy cold-drawn sheet, and the protection level reaches IP66;
- 8) Double-layer sealing protection for the box;
- 9) All electrical and mechanical components are protected against interference;
- 10) With functions of tap changer and transformer gas monitoring and online automatic oil filtration;
- 11) Automatic adjustment of grid voltage without AVR;
- 12) Parallel operation of multiple transformers without AVR;
- 13) Modular design, building block expansion, better resource allocation scheme;
- 14) High-precision transmission tapping control system;
- 15) With a variety of communication interfaces and communication protocols, provides more choices for smart grid applications.

**NOTE!**

Need to be in operation with tap changer. Special design is available to meet particular environment conditions. Please contact Huaming for further information.

4. Function introduction and operation conditions

- 1) The main function of the operating mechanism
 - a) Manual and Motor Operation;
 - b) Remote control, local control, CNC operation;
 - c) Step-by-step operation, overrunning at special request;
 - d) Local position indication, remote position indication and CNC position indication;
 - e) Tap position indication;
 - f) Electrical limit device;
 - g) Mechanical limit device
 - h) Local operation button;
 - i) Operation record;
 - j) Overload protection;
 - k) Parallel control device;
 - l) Restart Device;
 - m) Protection device against overstep;
 - n) Standard optical fiber transmission interface;
 - o) Over-current blocking signal input;
 - p) BCD signal output;
 - q) One-to-one contact signal output;
 - r) 4-20mA or 1-5V analog output signal.;
 - s) Operation status signal output;
 - t) Phase loss and phase sequence error of three-phase power supply signal output;
 - u) Timeout judgment signal ouput
- 2) Operating Mechanism Conditions
 - a) Ambient temperature -25 ~ 60°C; (special design can meet the ambient temperature of -60°C);
 - b) The vertical installation inclination shall not exceed 2%;
 - c) No serious dust and explosive or corrosive gas



NOTE!

Special design is available to meet particular environment conditions. Please contact Huaming for further information.

5. Parameters

Description		SHM-X
Motor	Rate power	0.75kW/1.1 kW/1.5 kW
	Rate voltage	110/220/380V 3 phase (can be customized)
	Rate current	3.48A/2.01A
	Rate frequency	50、60HZ
	Rated rotational speed	1400r/min
Rated torque of drive shaft (N·m)		18 (for HWDK) /35-65
Revolutions of drive shaft each step		33
Revolutions of hand crank each step		33
Automatic operation time of each step (S)		About 2.8 (for HWDK) /4.8
Maximum number of working positions		107
Insulation level kV (50Hz,1min)		2
Weight (kg)		85
heating loop circuit		220VAC
Heater power consumption		50W/100W/200W
Protection class of enclosure		IP66
Controller	Code	SHM-K、ETSZ-6等
	Rate voltage (V)	85 ~ 265V/AC
	Frequency (Hz)	50、60



NOTE!

The technical parameters listed in this manual are only applicable to standard designs, which may vary due to different ordering requirements, and are subject to change without notice.

Table 2 Function

Local Control Module		Tap position indication, regulation	
		Three-phase motor, Tap measurement sensor, optical fiber communication technology	
	Basic function	Electrical and mechanical limit protection, power failure protection	
		Phase sequence and phase loss detection of three-phase power input	
		变换超时保护 Transform Timeout Protection	
		Temperature and humidity control, heating plate control	
		Output	One channel of BCD code, RS485, MODBUS protocol
		Operation status (Dry contact)	Voltage regulation signal ((rise or lower), Max. operation position, Min. operation position, Incomplete signal, on-line oil filter control, hand crank operation signal, remote/local control signal
		Manual operation at controller	Optic electronic isolation, N/O contact input
		Temperature control	Available for severe cold temperature
		Interlocking	Remote or online monitoring signals
	Optional	Other signal output	Two sets of 4-20mA, 1-5V output One-to-one contact signal output (dry contact), Resistor signal output 2 sets of BCD signal output Other communication protocol besides Modbus
		Humidity adjustment	For area of high humidity
Remote control Module	Basic function	Remote position control	Tap position indication、operation
		Digital Communication	Method: UART serial communication Protocol: Modbus/CDT/IEC101/IEC103/1801
		Position output	BCD code
		Operation Historic record	Action time, tap position, fault record
		Local mode operation situation	Display all local panel operation status. Max. Min. position Working status local communication status, failure information, etc.
	Optional	Automatic voltage regulation	
		Multiple parallel operation	2-12 units
		Other position output	TCP/IP network Protocol: IEC104, etc. Others are the same as local mode


NOTE!

The technical parameters listed in this manual are only applicable to standard designs, which may vary due to different ordering requirements, and are subject to change without notice.

6. Vacuum circuit breaker detection controller

The vacuum circuit breaker detection controller (hereinafter referred to as the control module) is used to detect whether there is current after the vacuum circuit breaker is disconnected with charge.

1) Main parameter

Power supply: 115 ~ 230VAC/DC 15VA

Operating temperature and humidity: -25°C ~ 70°C, 95%RH

Dimension: 125 mm×108 mm×50mm

Storage temperature: -40°C ~ 85°C

2) Wiring installation

The controller is installed with DIN rail

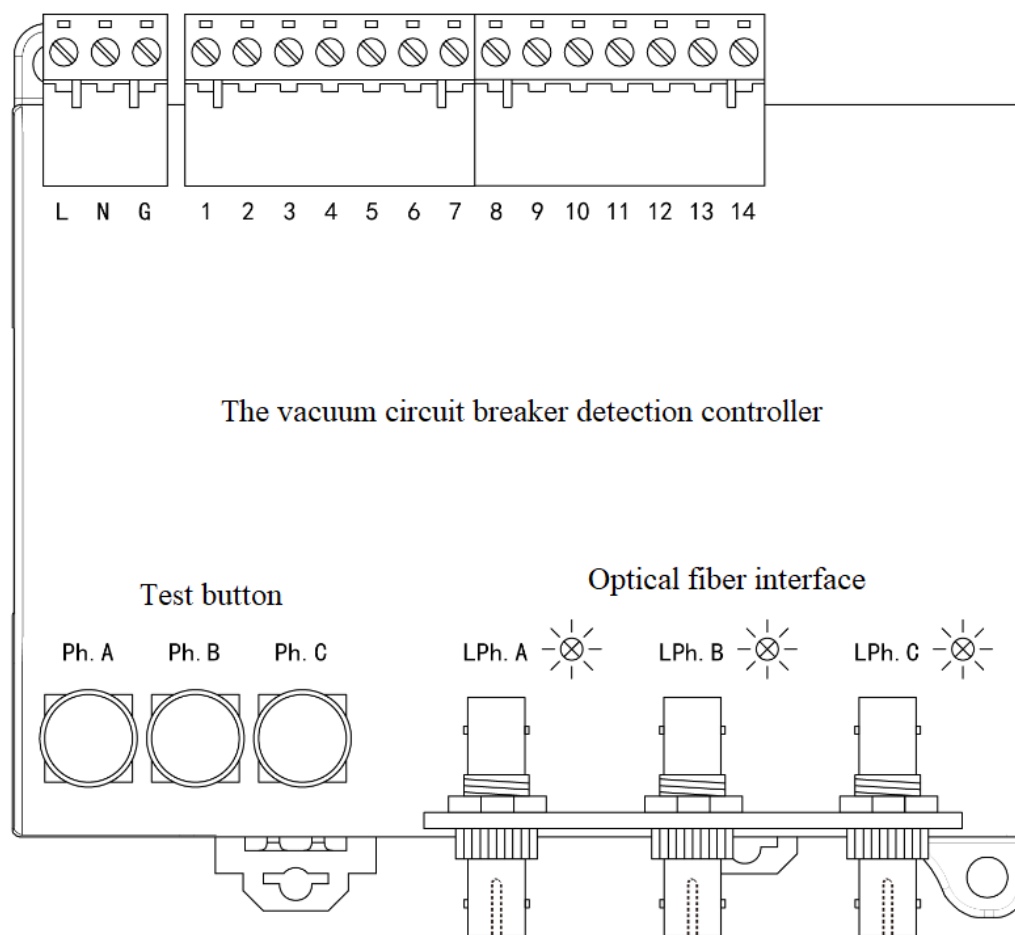


Fig 6-1

Terminal code	Terminal meaning	Remark
L, N	Respectively connect the positive and negative AC phase line, zero line or DC of the working power supply	115 ~ 230VAC/DC
G	Grounding	Protect grounding
1, 2	Cam signal connected to reactance switch (input signal)	Passive normally open contact (Input by external device)
3, 4, 5	Access upshift command, downshift command and common terminal (input signal)	
6, 7	Normal signal output of transformer	Passive normally open contact (Contact load capacity 5A/250V AC, 5A/30V DC)
8, 9	Locking signal output	
10, 11	Stop signal output	
12, 13, 14	Return upshift signal, return downshift signal and common terminal	
Optical fiber port LPh.A, LPh.B, LPh.C	Respectively connect reactance switch phase A, phase B and phase C current detectors	The optical fiber port is ST interface. Please see the notch clearly when installing the optical fiber port, and insert and fix it carefully
Test button Ph.A, Ph.B, Ph.C	Respectively point to phase A, phase B and phase C current detectors	

3) Function description

When the user use motor drive unit to switch On-load Tap changer, the control module collects and analyzes the operating condition of the mechanism and monitors whether there is residual current flowing after the vacuum circuit breaker is disconnected. If the monitored residual current is greater than a certain value set, the control module immediately sends a stop command to the motor drive unit, stopping the motor running (working) immediately. After the control module delays for hundreds of milliseconds, the control module performs the return operation according to the original running direction of the motor drive unit. The main purpose of the control module delay operation is to protect the motor from overheating damage due to reverse overshoot. The motor drive unit receives the return operation command and then performs the return operation. After the motor runs back to the starting position of this On-load Tap-changer operation after several seconds, the control module

sends a locking signal to the motor mechanism to lock it to prevent the user from operating the motor drive unit again until the fault is eliminated.

The three optical fibers are respectively connected to the optical generator of the current transformer and the optical receiver of the control module which next to the three vacuum interrupters (circuit breakers). When the vacuum circuit breaker is disconnected, if the optical receiver of the control module detects that the CT current transformer of one or more optical fibers exceeds the optical intensity of the set current value, the control module will immediately perform the above operations. At the same time, the position of the optical fiber that displays the lit overcurrent value is triggered, that is, the current transformer measured by a certain optical fiber sense that the vacuum degree of the vacuum circuit breaker has dropped or failed.

The control module monitors the CT operating conditions online. If the three CT operating conditions are consistent, the controller will output normal signals. If the three CT operating conditions are inconsistent, the controller will release the output abnormal signals.

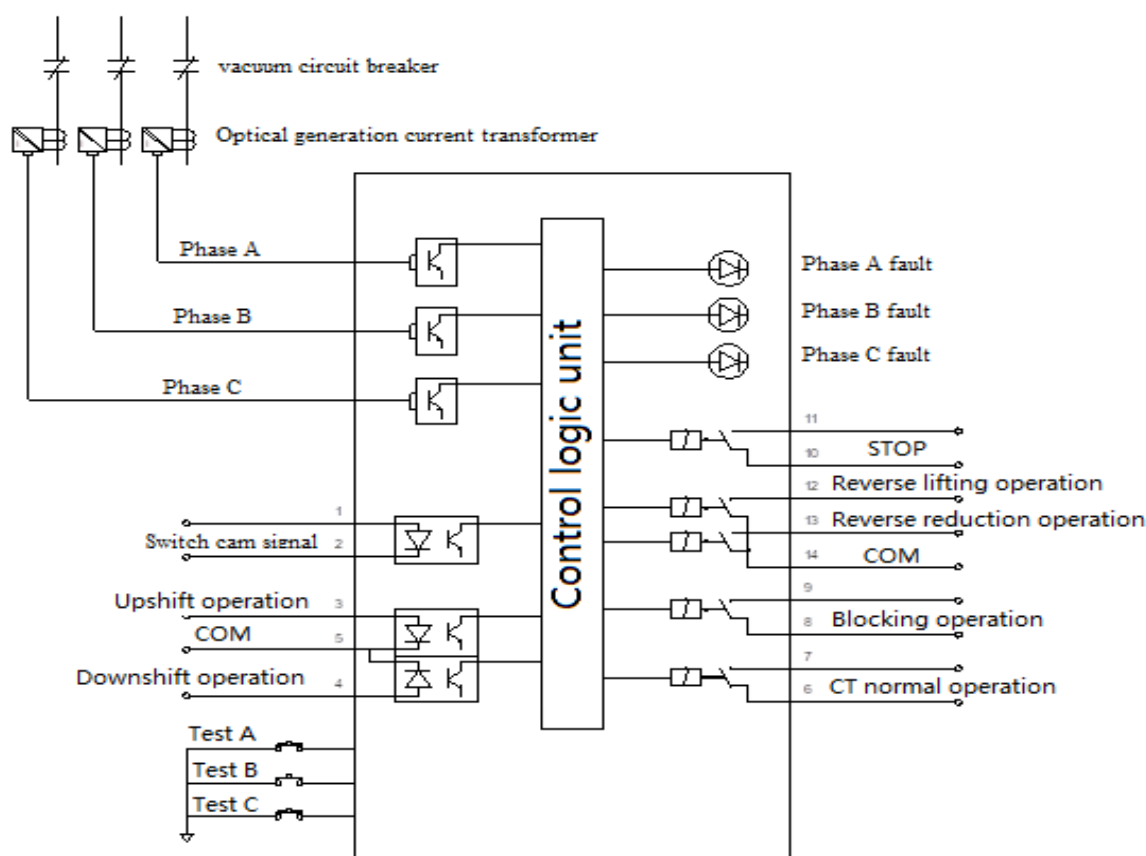


Fig 6-2 Principle of vacuum circuit breaker detection controller

4) Instructions for use of test button

1) When the motor mechanism is in the static state and the On-load Tap-changer is in the non-bridging position, press a test button, and the abnormal indicator of the control module will light up, indicating that this signal has changed and is different from other line signals.

2) When the motor drive unit is in operation, press a test button to simulate the failure of the vacuum circuit breaker. At this time, the control module detects the failure of a vacuum circuit breaker, immediately commands motor drive unit to stop running, and then orders the motor drive unit to run in the opposite direction of the original running direction until the motor drive unit returns to the start position. The line fault indicator lamp of the control module lights up and send the locking signal to the locking contact of the motor drive unit to prohibit the next operation of the motor drive unit.

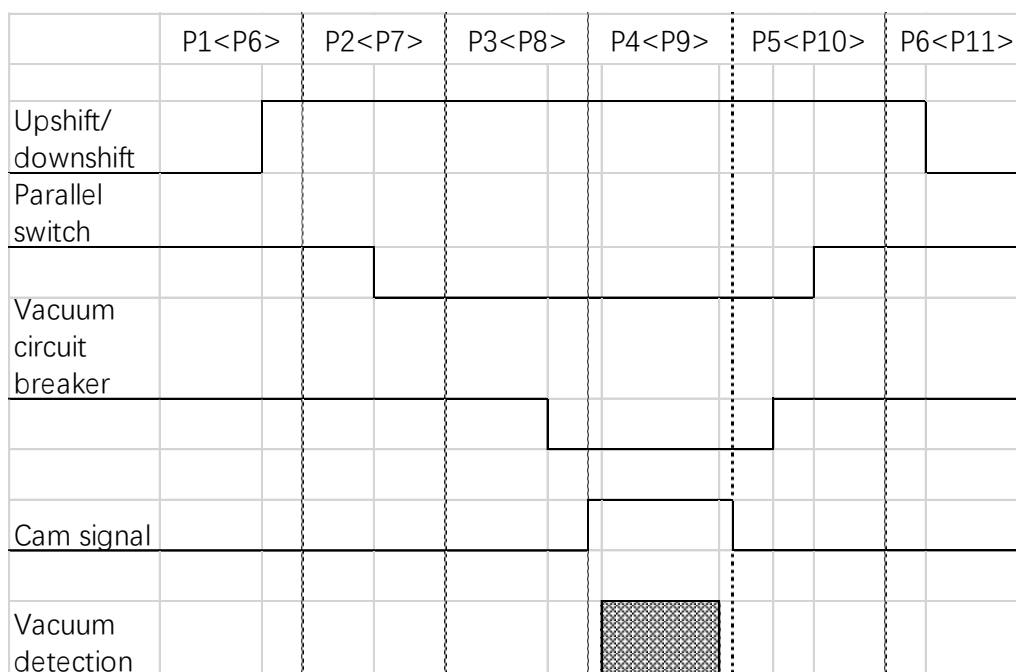


Fig 6-3 Requirements for working sequence of motor drive unit and reactance On-load Tap-changer

The control module detects whether there is vacuum leakage in the shaded part of the vacuum circuit breaker. If the current of the vacuum circuit breaker is greater than the CT setting value when it is opened, it indicates that the vacuum circuit breaker has the phenomenon of insulation decline and arcing. The control module sends the command to stop the operation of the motor drive unit and commands the motor drive unit to operate in the opposite direction of the original operation until it reaches the starting position. Then send the locking signal to the locking contact of the motor drive unit, which can prohibit the next operation of the motor drive unit.

It is recommended to use DC power supply for the control module to ensure reliable power supply. The vacuum circuit breaker must be disconnected before the cam signal appears, and can be closed only after the cam signal is released

7. Installation and Operation

7.1 MDU installation on transformer tank.

7.1.1 The mounting base thickness of MDU is 10mm, with installation hole 4×15mm and spacing see **Appendix 1**. Users shall prepare four M12 bolts and corresponding washers and nuts, length of bolts depends on the thickness of transformer mounting bracket.



NOTE!

Make sure that the mounting surface is flat so four mounting bases of the MDU will contact the mounting bracket at the same time.

To avoid the unevenness, please adjust with the gasket to ensure the gap is within 1mm.



NOTE!

It must be ensured that the serial number on the MDU nameplate is consistent with the serial number of the tap changer installed on the transformer before mounting.



NOTE!

If the transformer generates severe mechanical vibration during operation, please use a damping device.

7.1.2 The grounding position of the cabinet locates at the lower right corner, see **Appendix 1**, with bolt size M12×25. Users can adjust the grounding bolts to other three positions according to needs.



NOTE!

Grounding terminals of all electrical components and parts are equipotential connection with the cabinet, no extra grounding is required inside the cabinet.



WARNING!

It is not allowed to connect transformer's other device earthing wire to the MDU's grounding terminal.

7.2 Connection between MDU and tap changer

7.2.1 Along with the tap changer, there are two driving shafts, four sets of brackets (two for vertical shafts and two for horizontal shafts) and four pieces of driving shaft pins.

7.2.2 Maximum length of driving shaft is 2000mm, users can modify its length according to the actual distance between shaft ends of MDU and bevel gearbox as shown in Figure 7-1, needed length of driving shaft is $L-5\text{mm}$.

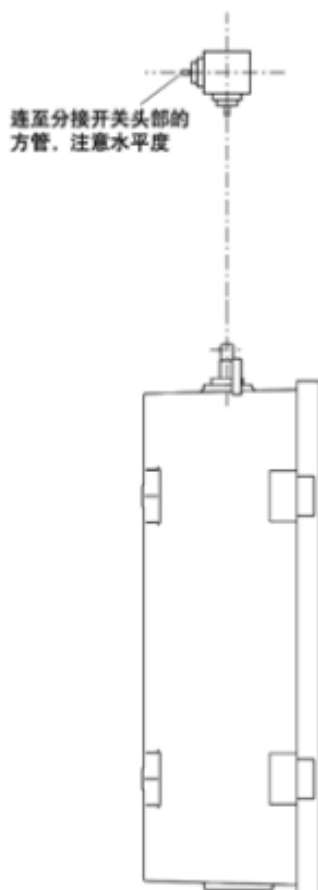


Fig 7-1

NOTE!



Before connecting MDU with tap changer, please make sure they are set at same position.

The perpendicularity of driving shaft should be less than 2%. Universal joint can be used when necessary.

7.3 Adjustment of operation cycle



WARNING!

Only manual operation is allowed and make sure the MDU and tap changer operate at the same tap position.

When hand crank is engaged, insert another 8mm (hand crank to the end) and make internal bevel gear engage, then crank the MDU. Push the hand crank hard during hand cranking to overcome the spring thrust.



NOTE!

Both MDU and tap changer shall be at setting position before this operation.

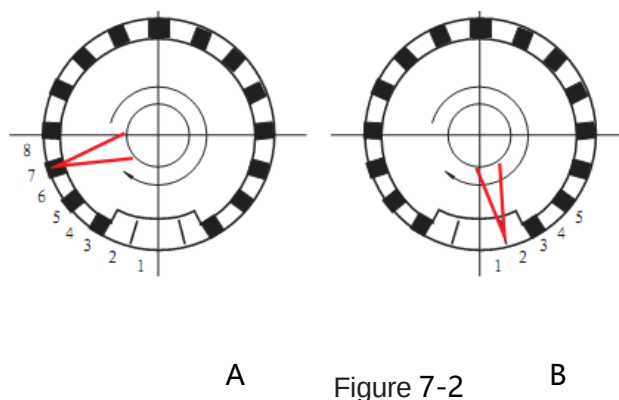
7.3.4 Case study

a) The tap changer is at position 10, operate the hand crank to position 11 till the diverter switch operates, record the pointer's location.

Result: 7 grids to 6 o'clock, like Figure 7-2. A.

b) The tap changer is at position 11, operate the hand crank to position 10 till the diverter switch operates, record the pointer's location.

Result: 1.5 grids to 6 o'clock, like Figure 7-2. B.



Adjust value: $(7-1.5)/2=2.75$, adjust ≈ 3 grids

Method:

- Disconnect the vertical shaft
- Operate the hand crank to position 12 for 3 grids
- Reconnect the vertical shaft
- Operate the hand crank to position 10, record the pointer's location:

Result: 4.5 grids to 6 o'clock

e) Check the counter direction's difference of position 11

Result: 4 grids to 6 o'clock

Adjustment done.

7.4 MDU operation

7.4.1 MDU power supply connects to terminals 3, 5 and 7 of X1, three-phase 380V 50/60Hz.

7.4.2 The MDU is connected with remote controller SHM-K with optical fiber. Plug the fiber into the CX interface, the MDU can realize remote control and operation status and information transmission via controller, Figure 7-3 and 7-4.

Figure 7-3 MDU CX Interface

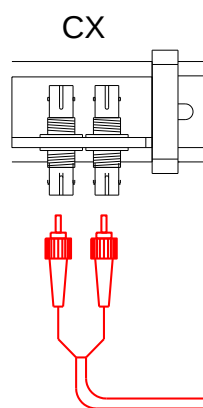
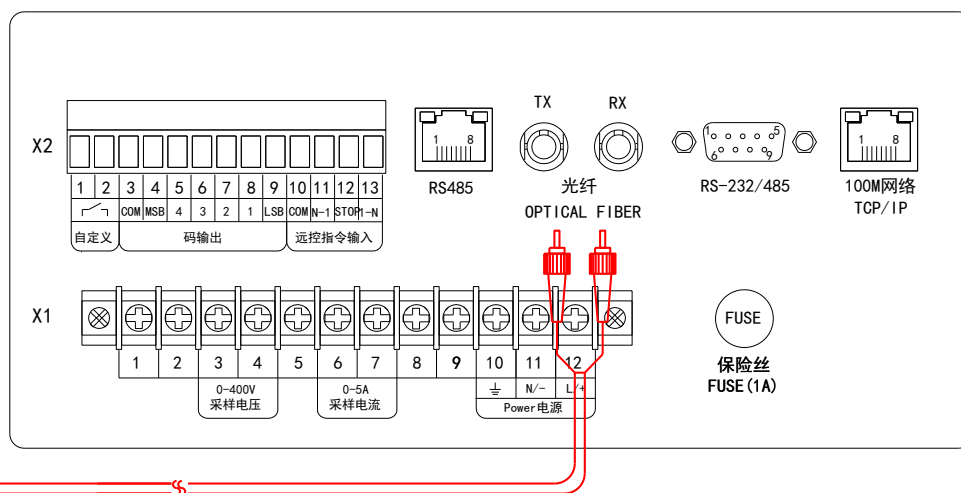


Figure 7-4 SHM-K back Interface



WARNING!

Minimum bend radius for optical fiber during wiring or configuration is R30 mm.

NOTE!

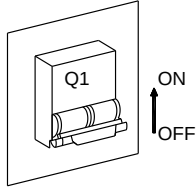
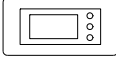
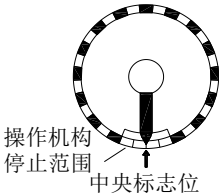
The optical fiber inlet entrance locates at the cabinet bottom as shown in [Appendix 1](#), with diameter 20.5mm, permanent joint is along with the fiber.

During the commission, test and when automatic or remote control is not required, MDU can operate separately without SHM-K. Connect power supply as per 6.4.1 and select local mode, then operate the MDU.

Before operating the MDU, please read the operation instruction carefully especially [Appendix 3, 4, 5](#) and 6.4.3 to understand the function and operation method of above-mentioned electrical components.



7.4.3 MDU operation instruction

1	Turn on circuit breaker Q2 and energize local control panel. Then Nixie tube on local control module firstly displays configuration code and current tap position, confirm whether the tap is correct.	
2	Turn on three-phase circuit breaker and energize three-phase asynchronous motor.	
3	Select proper mode by the buttons on SHM-X local control module. When select  'Local' mode (local button control), it is controlled by 'raise', 'lower' and 'stop' buttons on the local control module panel. When select  'Communication' mode (remote numerical control), it is controlled by SHM-K remote controller via optical fiber and realize status and information transmission. When select  'Remote Control' mode (remote button control), remote control input terminal is provided inside MDU and realize control by external signal via terminal.	 远方键控  远方数控  就地键控
4	The control buttons for local mode locate on the SHM-X local module. Press  the MDU raises one position. Press  the MDU lowers one position. Press  the MDU stops operation immediately and breaks three-phase circuit breaker.	  
5	After each 'raise' or 'lower' operation, the changing status pointer shall stop within “MDU stop range”. The “central marked position” is used for calibrating the accuracy of MDU operation.	
6	Three-phase circuit breaker will be cut off by one of hand crank insert, electrical limit or any 'stop' order.	


WARNING!

When perform high voltage test for the transformer, please remove MDU's external wiring to avoid damages to components inside the MDU.

7.5 Transportation

NOTICE!



During transformer transportation without MDU, the reinstallation shall follow instruction 7.1~7.4.

When the transformer is transported by ocean (salt fog or hygrothermal environment), please wrap the MDU and driving shaft with plastic film (or plastic bag) to avoid long time exposure to the open air.

7.6 Maintenance

During the routine maintenance of the transformer, please carry out following inspections on the MDU:

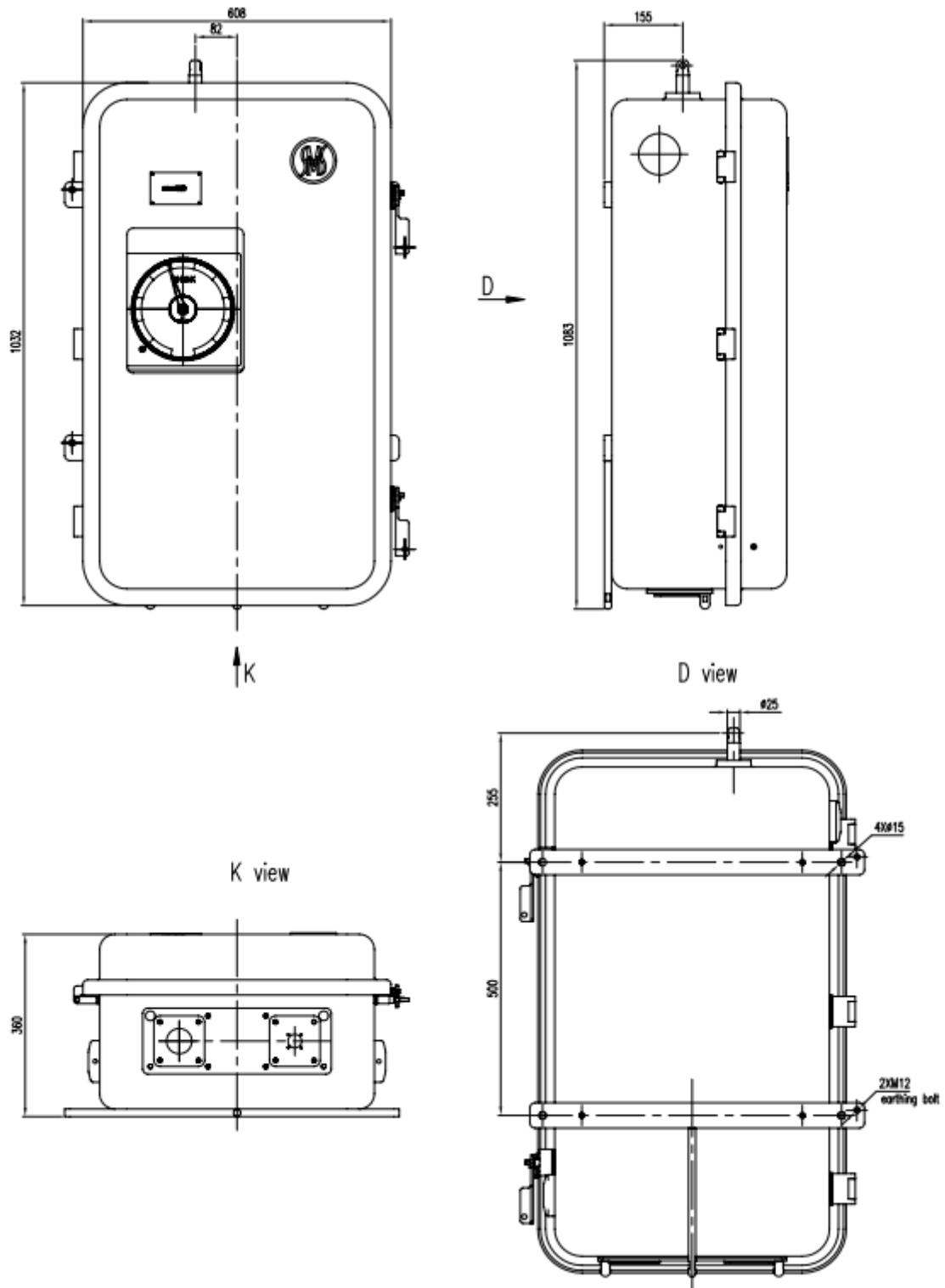
Check the sealing elements of the MDU cabinet to ensure reliable sealing performance.

Check the appearance status of the components inside MDU.

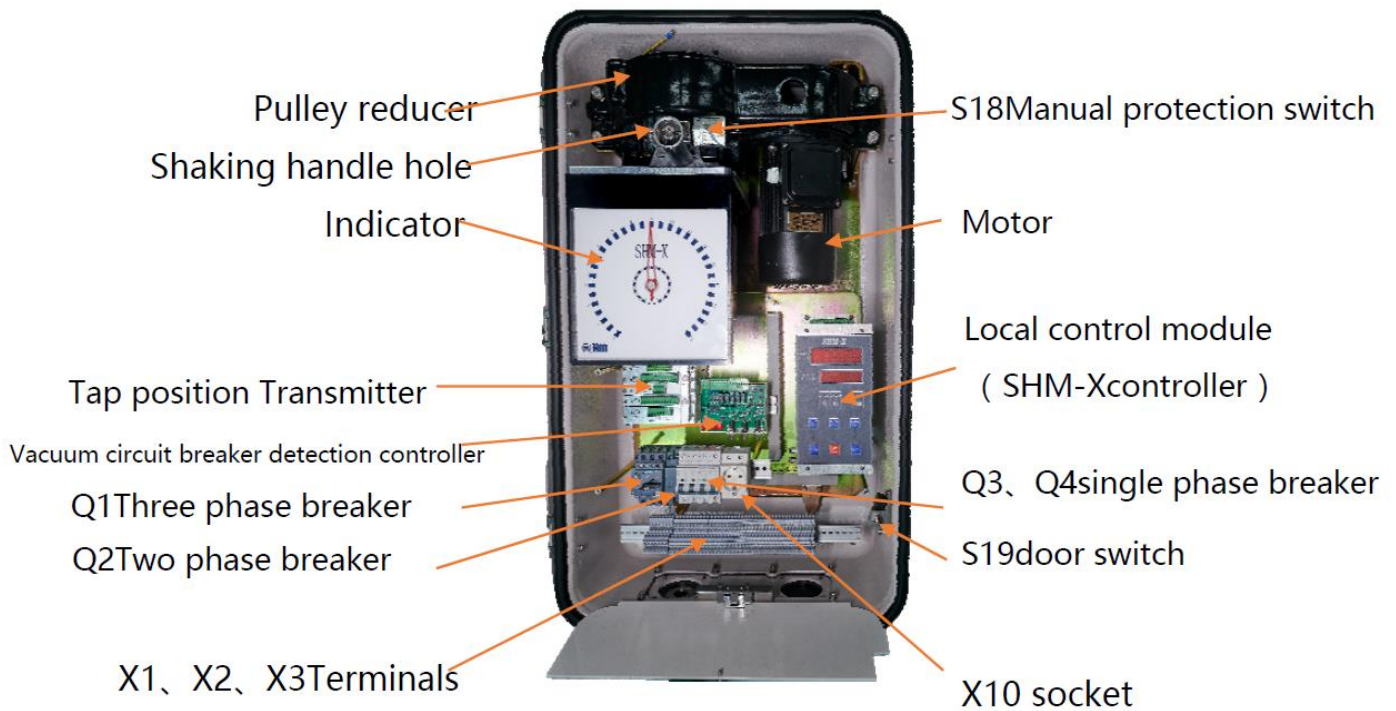
Controller error alarm

Error Code	Fault Cause
E-01	Hall sensor error
E-02	Configuration information error
E-03	Motor rotation direction
E-04	Tapping sensor error
E-05	Three-phase power supply circuit breaker control error
E-06	Temperature and humidity sensor error
E-07	Memory stick error
E-08	Memory monitoring error
E-09	Default phase
E-10	Power Q1 input error

Appendix 1 SHM-X outline drawing



Appendix 2 Internal components

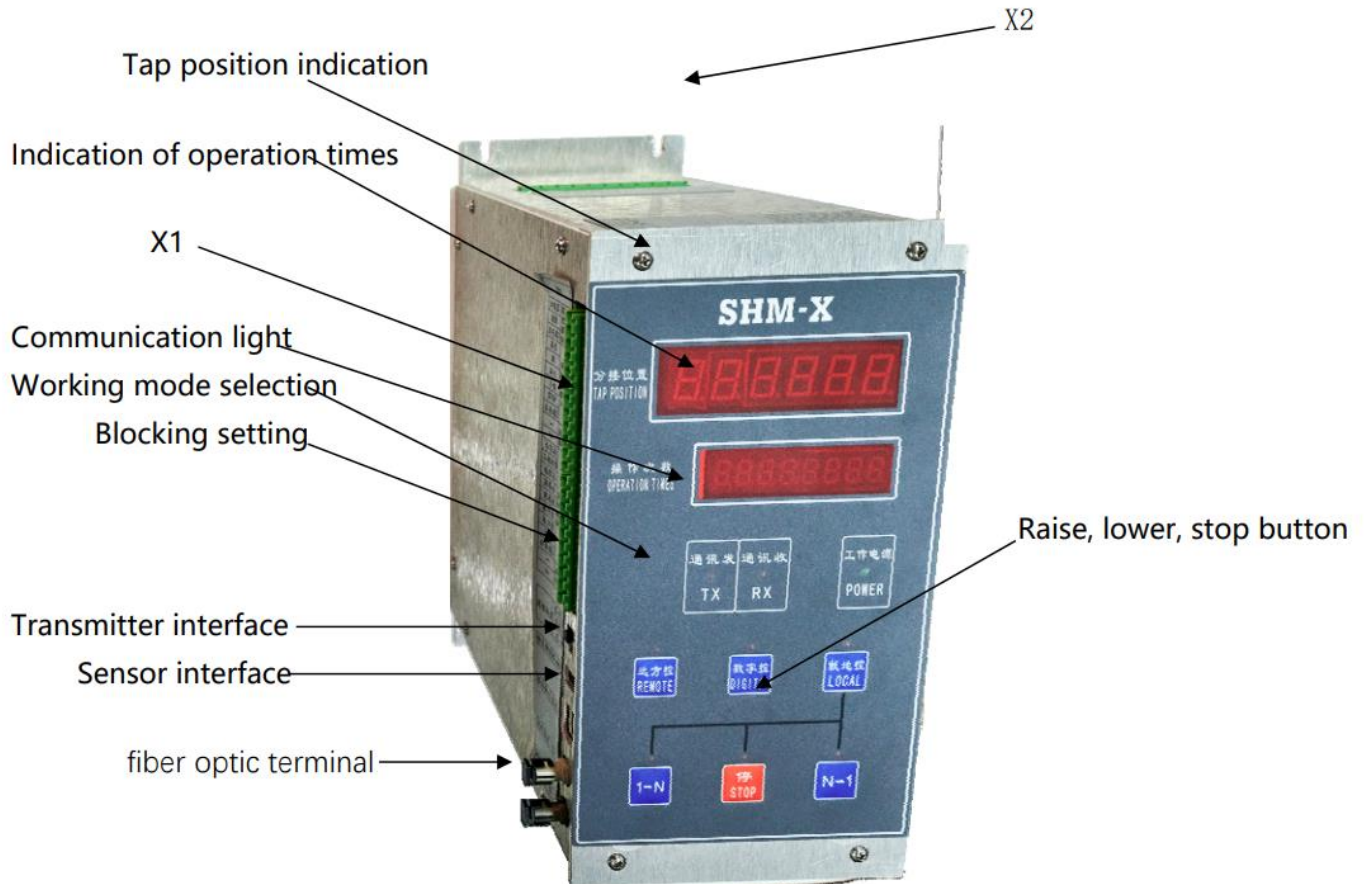


NOTICE!



1. The SHM-X operating mechanism and the remote controller SHM-K are connected by optical fiber, and the supporting optical fiber is inserted into the optical fiber interface and connected, and the remote control and switch operation status information transmission can be realized through the controller.
2. When the optical fiber is connected, the two communication indicators of the local control module CH2 in the operating mechanism flash to indicate that the connection is correct; otherwise, the connection is faulty. Please exchange the two optical fiber connection plugs.
3. An overcurrent blocking contact input terminal is provided in the motor mechanism. By default, the input is a N/O contact or a N/C contact (the Locking contacts must be consistent, please refer to Appendix 4). When the N/O contact is closed or the When the N/C contact is opened, the operating mechanism is locked.

Appendix 3 Schematic diagram of local control module

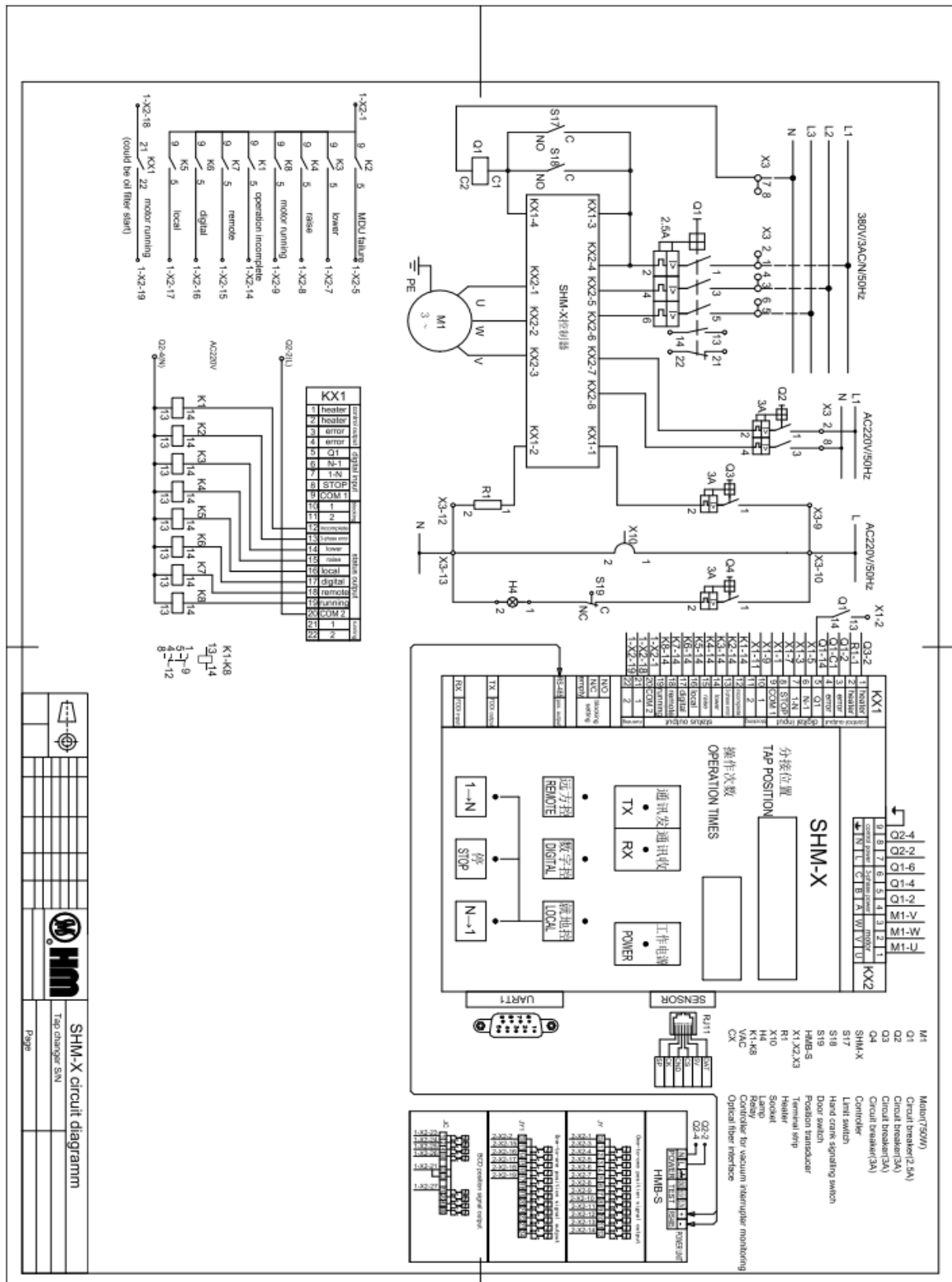


	1	2	3	4	5	6	7	8	9
X2	Motor			Three-phase power			Control power		
	U	V	W	A	B	C	L	N	

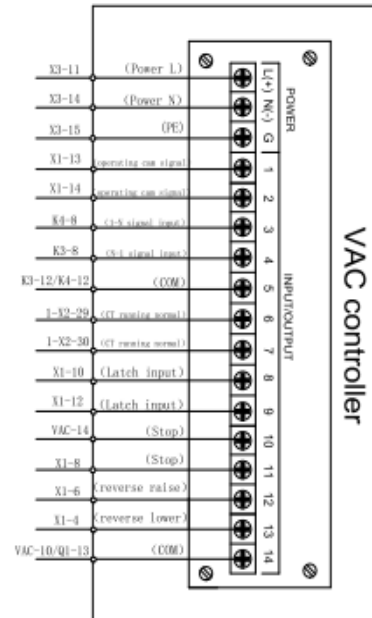
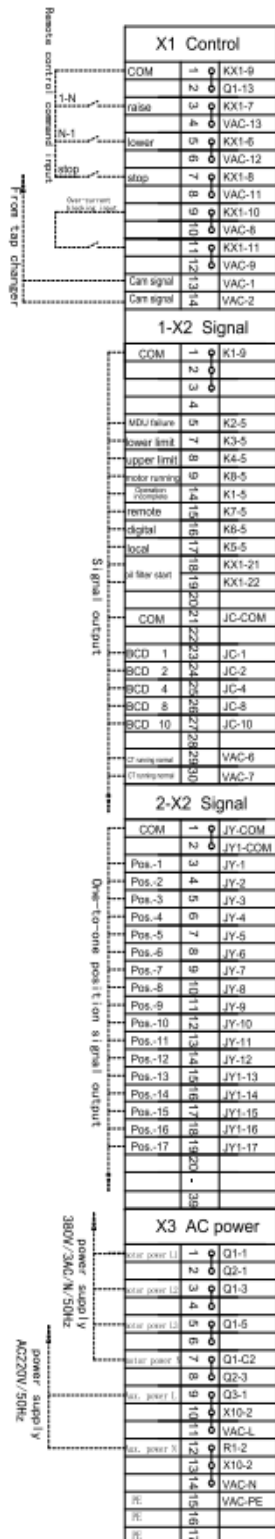
X1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
	Output control				Input					Blocking		Output									Operation	
	Heater		Fault		Q1	1 — N	N — 1	S T O P	Common terminal 1	1	2	Un-finish	Fault	Raise	Lower	Local	CNC mode	Remote	Inoperation	Common terminal2	1	2
	1	2	1	2																		

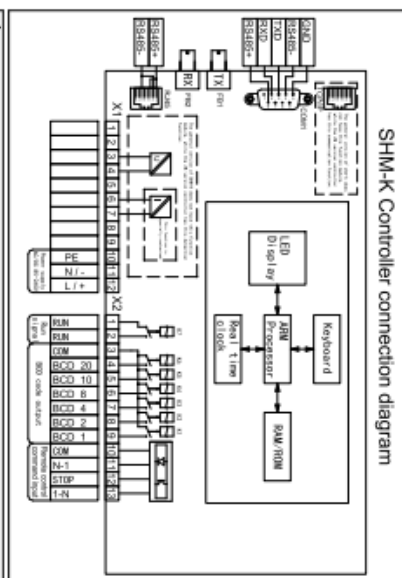
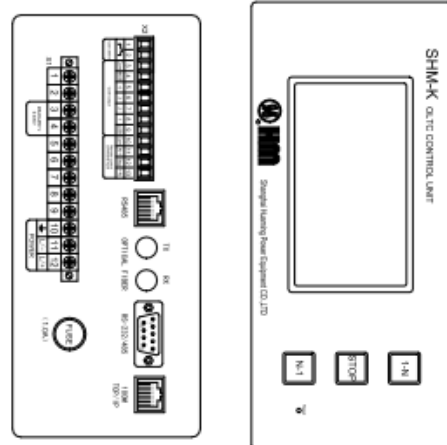
Blocking setting		
N/O	N/C	N/A

Appendix 4 Electrical schematic Diagram

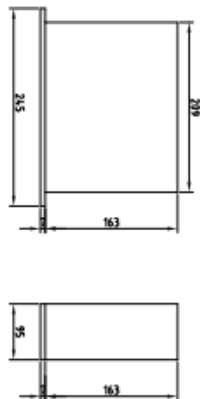


- NOTE:
1. SHM-X need power supply inside the cabinet, 380V 50Hz power supply at No. 1,3,5,7 of terminal X3.
 2. 220V 50Hz at No. 9,12 of terminal X3.
 3. When contacts close, MDU will be blocked, N/O is factory setting.
 4. Contact capacity: DC220V 0.5A / AC250V 5A.

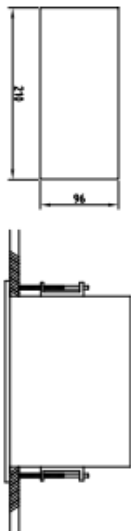




⚠ SHARK controller XT terminal block 10th ground terminal must be grounded.



Installation diagram



Dimension diagram

Units:mm

[illegible]

